

Efficient and Stable Electrocatalytic Oxygen Evolution from $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ Heterostructures

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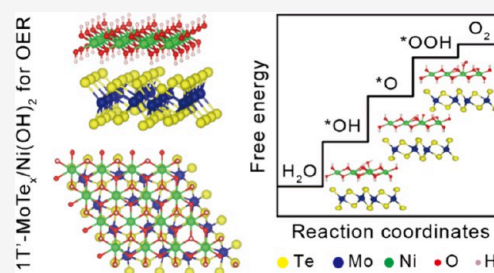


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ABSTRACT: A design methodology is presented here for $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructured catalysts that enhance the oxygen evolution reaction (OER) in water electrolysis. This approach relies on the transfer of mechanically exfoliated MoTe_2 nanosheets to Au/Si substrates followed by electrochemical Te dissolution to induce defect-mediated, partial semiconducting (2H) to metallic (1T') phase transitions. Immersing the resulting MoTe_x into a nickel nitrate hydrate solution results in a heterostructure consisting of 2H- $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ and 1T'- $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ domains, which enables high stability and improved efficiency for the OER compared to IrO_x . Both machine-learning potential and density functional theory calculations searched and evaluated all atomic sites for this materials system, thus revealing the enhancement mechanisms of OER via four-electron transfer processes with lowered free energy barriers in the rate-determining steps.



INTRODUCTION

The development of earth-abundant, efficient, and stable electrocatalysts for the oxygen evolution reaction (OER) remains one of the most important research topics in applied electrochemistry for energy conversion and storage systems such as water electrolyzers and metal–air batteries.^{1–5} In comparison with the cathodic hydrogen evolution reaction (HER), anodic OER has sluggish kinetics due to a large energy barrier for water splitting since it involves four sequential proton-coupled electron transfer processes and the formation of O=O bonds.^{6,7} Nonoxide-based transition metal compounds, particularly transition metal dichalcogenides (TMDs) with higher electrical conductivity than transition metal oxides, have shown significant promise as efficient OER catalysts for alkaline water splitting to replace electrocatalysts based on expensive noble metals such as iridium (Ir)- and ruthenium (Ru)-based oxides.⁸ The stability of TMDs in OER, however, is problematic because these materials undergo *in situ* electrochemical transitions under anodic oxidation conditions, resulting in deleterious conversion to transition metal (oxy)-hydroxides.^{9–11}

The tunable growth of metal hydroxides on the “pre-catalysts” of TMDs has enabled design and synthesis of hybrid electrocatalysts for stable water electrolysis.^{7,9,12} For example, the formation of Ni, Co, Fe, and Mn hydroxides on MoS_2 and WS_2 has improved alkaline HER due to the effective dissociation of water molecules (required for the initial Volmer step in HER) at heterointerfaces.^{13,14} In particular, heterostructures consisting of $\text{Ni}(\text{OH})_2$ nanoparticles and MoS_2

nanosheets have achieved OER with reduced barrier energies for the formation of intermediate species (i.e., $^*\text{OH}$, $^*\text{O}$, and $^*\text{OOH}$) during the four-electron transfer steps.¹⁵ Most of these hybrid electrocatalysts based on MoS_2 , however, have exhibited lower OER performance than Ir- or Ru-based oxides, and the stability of electrochemical reactions has been limited to less than 100 h.^{15–23} Furthermore, fundamental investigations to reveal the mechanisms regarding how the crystal structures of TMDs (e.g., semiconducting 2H and metallic 1T phases) and the formation of chalcogen defects affect the resulting OER are still lacking, which has limited the establishment of rational design rules for synthesizing electrocatalysts containing such heterointerfaces.

RESULTS AND DISCUSSION

Here, we report the formation of $\text{Ni}(\text{OH})_2$ domains on defect-engineered MoTe_x crystals to construct heterostructures having an OER efficiency and stability far greater than those of iridium oxide (IrO_x). The dissolution of Te atoms from mechanically exfoliated MoTe_2 on gold thin films induces partial 2H-to-1T' phase transitions within the flakes. By immersion of the resulting heterophase MoTe_x into a nickel

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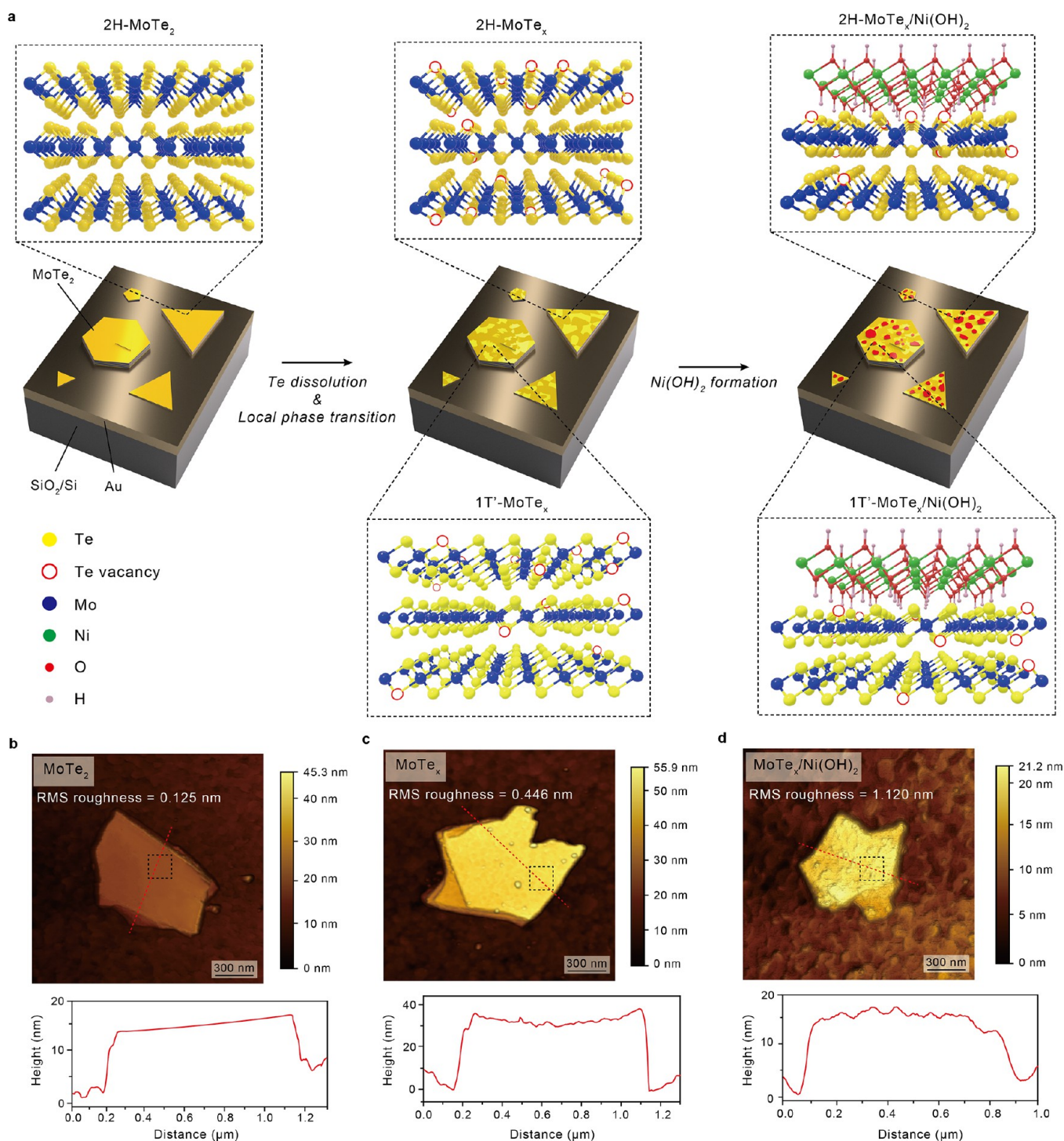


Figure 1. Fabrication of hybrid electrocatalysts based on mechanically exfoliated bulk MoTe_2 . (a) Schematics depicting the formation of $\text{MoTe}_x/\text{Ni(OH)}_2$ heterostructures. AFM images and corresponding cross-sectional profiles of (b) MoTe_2 , (c) MoTe_x , and (d) $\text{MoTe}_x/\text{Ni(OH)}_2$.

nitrate hydrate solution, Ni ions adsorb at the MoTe_x defect sites, after which electrochemically active Ni(OH)_2 nanoparticles nucleate. Correlated functional imaging based on Raman scattering, X-ray photoelectron spectroscopy (XPS), and transmission electron microscopy (TEM) reveals the formation of Ni(OH)_2 over the domain areas of 2H, 1T', and 2H/1T' boundaries. These $\text{MoTe}_x/\text{Ni(OH)}_2$ heterostructures show superior OER performance (with an overpotential of 180 mV for 0.1 mA/cm^2 and a Tafel slope of 38 mV/dec) with greater steady-state activity than IrO_x in addition to stability

for more than 150 h. Atomistic simulations employing machine-learning potential and density functional theory calculations provide insight into how these heterostructures benefit atomic-scale reactivities of electron transfer processes by lowering the formation energies of successive intermediates in the OER.

Figure 1a depicts the electrochemical processes for as-transferred MoTe_x flakes that give rise to the formation of $\text{MoTe}_x/\text{Ni(OH)}_2$ heterostructures based on distinct phase domains and interfaces. We first transfer multilayered flakes

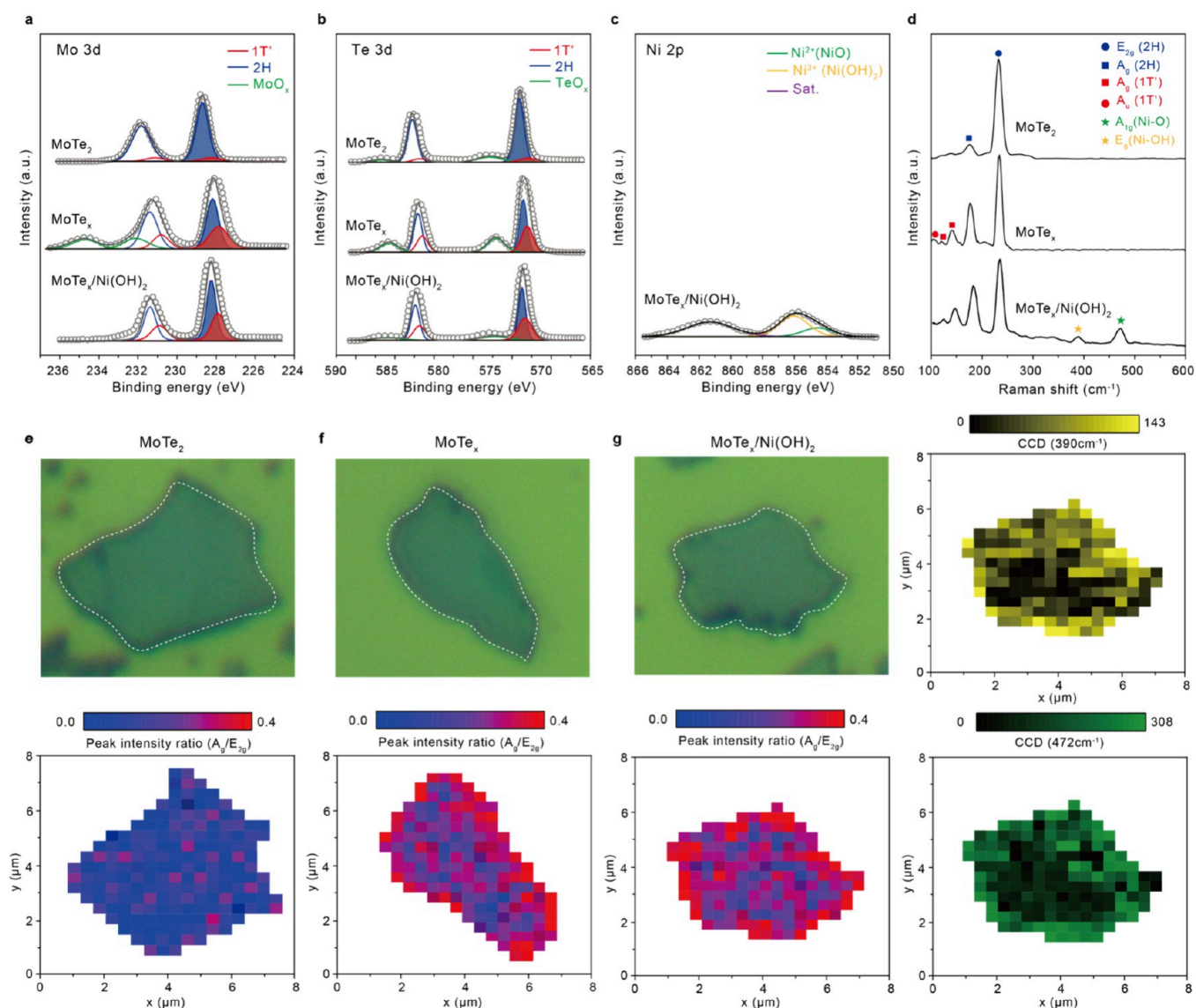


Figure 2. Spectroscopic characterization of MoTe_x/Ni(OH)₂. XPS (a) Mo 3d, (b) Te 3d, and (c) Ni 2p peaks for MoTe₂, MoTe_x, and MoTe_x/Ni(OH)₂. (d) Raman spectra of MoTe₂, MoTe_x, and MoTe_x/Ni(OH)₂. Spatially resolved phase transition from a single-flake-level measurement of MoTe₂ (e) before and (f) after Te dissolution and (g) Ni(OH)₂ formation verified by mapping the Raman peak intensity ratio of A_g (near 138 cm⁻¹, 1T' phase) to E_{2g} (near 228 cm⁻¹, 2H phase). The formation of Ni(OH)₂ on MoTe_x was also resolved by mapping A_{1g} (near 389 cm⁻¹, dehydrated Ni(OH)₂) and E_g (near 479 cm⁻¹, Ni(OH)₂) peaks. In panels (e)–(g), all Raman signals from other flakes and debris in the optical image aside from the flake enclosed by the dotted lines are excluded for clarity.

mechanically exfoliated from mineral MoTe₂ crystals onto Au/SiO₂/Si substrates (with an average surface coverage of ~39%) in a stepwise manner to achieve sample-to-sample consistency (Methods, Figures S1 and S2). Statistical analysis of the thickness and size distributions of the as-transferred MoTe₂ showed that flakes with an average thickness of ~18 nm (i.e., ~30 layers of MoTe₂) and lateral length of ~1.2 μm are transferred onto the Au films (Figure S3). We then connect copper wires to the Au films to fabricate working electrodes (WEs) for the OER, in a three-electrode configuration consisting of a Pt foil counter electrode (CE) and a Hg/HgO reference electrode (RE) with an alkaline electrolyte of 1 M KOH (Figure S4, Methods). By applying reductive cyclic voltammetry (CV) over the WE, we form Te defects on the basal planes of MoTe₂ flakes via electrochemical Te dissolution (Methods).^{24–26} These incorporated chalcogen defects promote local 2H-to-1T' phase transitions and the formation

of heterophase boundaries.^{27–31} After Te dissolution, the WE was removed from the electrochemical cell, and the MoTe_x flakes were floated onto a nickel nitrate hydrate solution to form MoTe_x/Ni(OH)₂ electrocatalysts (Methods, Figure S2). Atomic force microscopy (AFM) images in Figure 1b–d show that the flat surfaces of MoTe₂ flakes are transformed to 3D corrugated structures decorated with Ni(OH)₂ domains with the level of corrugation increased from root-mean-square (RMS) roughness of ~0.13 nm (as-transferred MoTe₂) to ~0.45 nm (MoTe_x) to ~1.1 nm (after Ni(OH)₂ formation). We verified that the size and thickness distributions of the heterostructures remained statistically consistent across arbitrary samples, demonstrating sample-to-sample reproducibility (Figures S5 and S6, Table S1).

Figure 2 and Figure S7 illustrate correlated XPS and Raman scattering analyses that reveal the transformation of MoTe₂ flakes into MoTe_x/Ni(OH)₂ heterostructures. As we reported

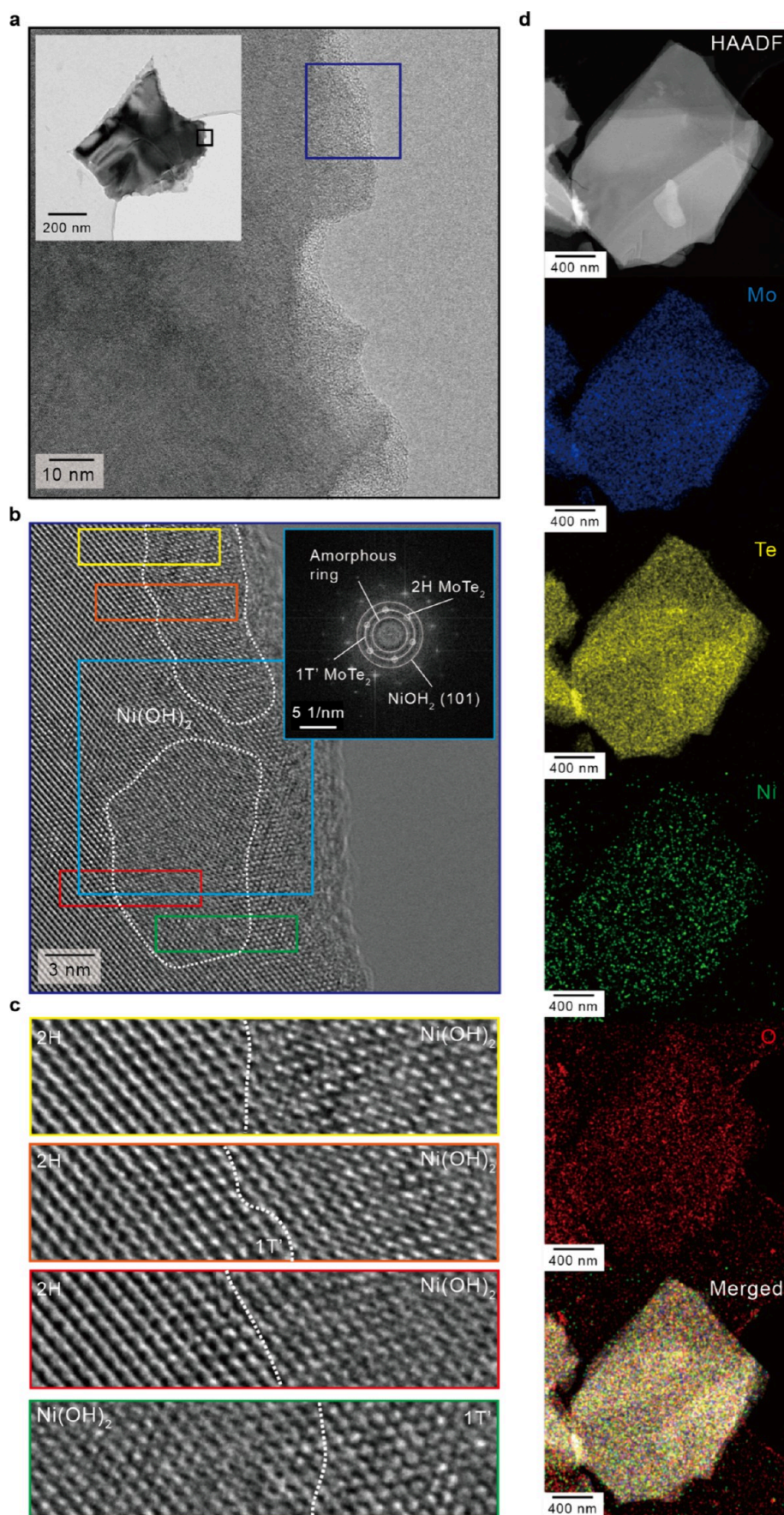


Figure 3. TEM analysis. (a) TEM and (b, c) HRTEM images of $\text{MoTe}_x/\text{Ni(OH)}_2$ near edge areas of the flake. (c) Compositional maps of $\text{MoTe}_x/\text{Ni(OH)}_2$. The HRTEM images in (c) show the heterointerfaces with $2\text{H-MoTe}_x/\text{Ni(OH)}_2$ (yellow and red boxes), $2\text{H-MoTe}_x/1\text{T}'\text{-MoTe}_x$ /

Figure 3. continued

Ni(OH)₂ (orange box), and 1T'-MoTe_x/Ni(OH)₂ (green box), all magnified from the images in panel (b). The FFT pattern in panel (b) was converted from the real space information over the local area marked with cyan.

previously, the as-transferred flakes are predominantly in the 2H phase, with a small amount of 1T' phase (with a 1T'/2H ratio of ~0.12) after the processes of mechanical exfoliation and hot pressing (Figure S8).³² The Mo 3d core level spectrum of as-transferred MoTe₂ flakes consists of two doublets at 228.7 and 228.1 eV and 231.8 and 231.1 eV, associated with 2H and 1T' phases, respectively (Figure 2a).³² Similarly, the Te 3d core level exhibits peaks at 573.3 and 583.7 eV, which corresponds to the 2H phase and the signatures of 1T' phase at 572.4 and 582.9 eV (Figure 2b).³² Upon electrochemical Te dissolution, the 1T'/2H phase ratio increases significantly to ~0.68, as observed by a decrease (increase) in the portion of 2H (1T') component in the deconvoluted spectra, since the incorporation of Te vacancies generates local internal strain to promote the formation of 1T' (Figure 2a,b and Figure S8).^{31,33–35} We note that CV causes ~8% of Te dissolution – i.e., the dissolutions of Te atoms from the flakes into the electrolyte (Methods). Upon Te dissolution, we observe the appearance of additional components at 232.3 and 234.9 eV in the Mo 3d spectrum originating from the formation of Mo⁶⁺ species (e.g., MoO_x), and a similar trend for the Te 3d spectrum where the peaks at 575.7 and 587.1 eV indicate the presence of TeO_x; the presence of broken Mo–Te bonds by Te dissolution favors oxidation of both Mo and Te atoms upon exposure to air.^{36,37} Since we performed the XPS measurements after taking the sample out from the aqueous media, the formation of MoO_x (and TeO_x) species is expected.

Upon immersing the MoTe_x in nickel nitrate hydrate solution, we observed additional peaks in the survey spectrum signifying the presence of Ni-related species, particularly the formation of Ni(OH)₂ while preserving the 1T'/2H ratio of ~0.67 (Figure 2c and Figure S8). Specifically, the Ni 2p core-level spectrum suggests the presence of NiO_x species (e.g., dehydrated Ni(OH)₂) at 854.8 eV, as well as Ni(OH)₂ at 856.3 eV^{18,38,39} with a quantified compositional ratio of 13.6% for (dehydrated) Ni(OH)₂ with respect to the MoTe_x support (Figure S8). As the samples were only briefly exposed to air during the change of aqueous media (Methods), we expect negligible oxidation of the exposed Mo and Te bonds of MoTe_x in agreement with the absence of MoO_x or TeO_x signatures in the Mo 3d and Te 3d core-level spectra, respectively. We highlight that the incorporation of Te vacancies by reductive CV was critical to form the heterostructures with Ni(OH)₂ since Ni ions tend to adsorb preferentially on the reactive sites (e.g., Mo and Te dangling bonds) of the MoTe_x support.¹⁸ On the other hand, without the presence of engineered chalcogen defects, we only observe a small amount of Ni(OH)₂ with the resulting relative ratio of 1.97% (Figures S9 and S10).

Raman scattering measurements further elucidate the formation procedure of MoTe_x/Ni(OH)₂ heterostructures (Figure 2d). Raman spectra of as-transferred MoTe₂ flakes exhibited vibrational modes at 175.3 and 230.9 cm⁻¹ associated with the 2H phase. Although the portion of 1T' with respect to 2H is very small with negligible mode volume, we detected a weak signal from this metallic phase at 139.2 cm⁻¹ that corresponds to the A_{1g} vibrational mode (Figure

S11).³² Upon electrochemical Te dissolution of MoTe₂, we clearly observe the appearance of vibrational modes originating from the 1T' phase at 102.8 and 120.4 cm⁻¹ in addition to increasing A_{1g} peak intensity at 139.2 cm⁻¹, which further verifies the coexistence of 2H and 1T' phases. When the sample is finally immersed into the nickel nitrate hydrate solution, the A_{1g} vibrational mode (389 cm⁻¹) from NiO and E_g vibrational mode (479 cm⁻¹) from Ni(OH)₂ were detected, suggesting the formation of Ni-related species.⁴⁰ We note that the signals from MoO_x and TeO_x were negligible, which indicates that these oxide species are mostly amorphous rather than having well-defined crystalline structure.^{36,37}

To understand the spatial distributions of the two MoTe_x phases as well as Ni(OH)₂, we performed Raman mapping for the corresponding structures. These maps were constructed by taking the intensity ratio of A_g (140.4 cm⁻¹) and E_{2g} (231.2 cm⁻¹) modes of the 2H and 1T' phases, respectively. Figure 2e–g reveals that electrochemical dissolution promotes the 2H-to-1T' phase transformations throughout the flakes, and the resulting heterophase surface was preserved even after forming Ni(OH)₂. Raman maps were also constructed from the intensity of the A_{1g} (389 cm⁻¹) and E_g (479 cm⁻¹) modes, which confirms the spatial distributions of (dehydrated) Ni(OH)₂. We note that the Raman signals associated with Ni(OH)₂ were much smaller on as-transferred MoTe₂ than CV-reduced MoTe_x, which is in agreement with the aforementioned XPS analyses (Figure S8–S10 and S12).

In order to resolve the atomic arrangements and chemical compositions of the samples, we performed transmission electron microscopy (TEM) analyses (Figure 3 and Figures S13 and S14). High-resolution TEM (HRTEM) images of the as-transferred sample show atomic fringes, and their corresponding fast Fourier transform (FFT) patterns indicate that the sample is predominantly in the 2H phase with locally included 1T' domains near the flake edges (Figure S13). Upon electrochemical Te dissolution, we observe that the metallic phase (1T') domains were distributed randomly throughout the flakes giving rise to 2H-1T' boundaries (Figure S13), the observations of which are further supported by the aforementioned XPS and Raman analyses. The MoTe_x/Ni(OH)₂ heterostructures in Figure 3a–c exhibit several distinct features in comparison to (as-transferred) MoTe₂ and MoTe_x: (1) multiple phase domains and heterointerfaces among 2H-MoTe₂, 1T' MoTe₂, and Ni(OH)₂; (2) local disorder within and throughout the flakes induced by Te dissolution; and (3) local amorphization of the MoTe₂ lattice. Considering the average thickness of the layered flakes (~18 nm), not only the fact that the Ni(OH)₂ domains are too thin but also the presence of the underlying MoTe₂ complicated the analysis of the orientational relationships between Ni(OH)₂ and the MoTe₂ lattice. However, the ring patterns from FFT associated with Ni(OH)₂ confirm the presence of the metal hydroxides and their polycrystalline nature. Energy-dispersive X-ray (EDX) maps in Figure 3d also suggest that the formation of Ni(OH)₂ was rather homogeneous and uniform over the entire surface of the flakes, which agrees with the corrugation observed in the AFM topography measurements.

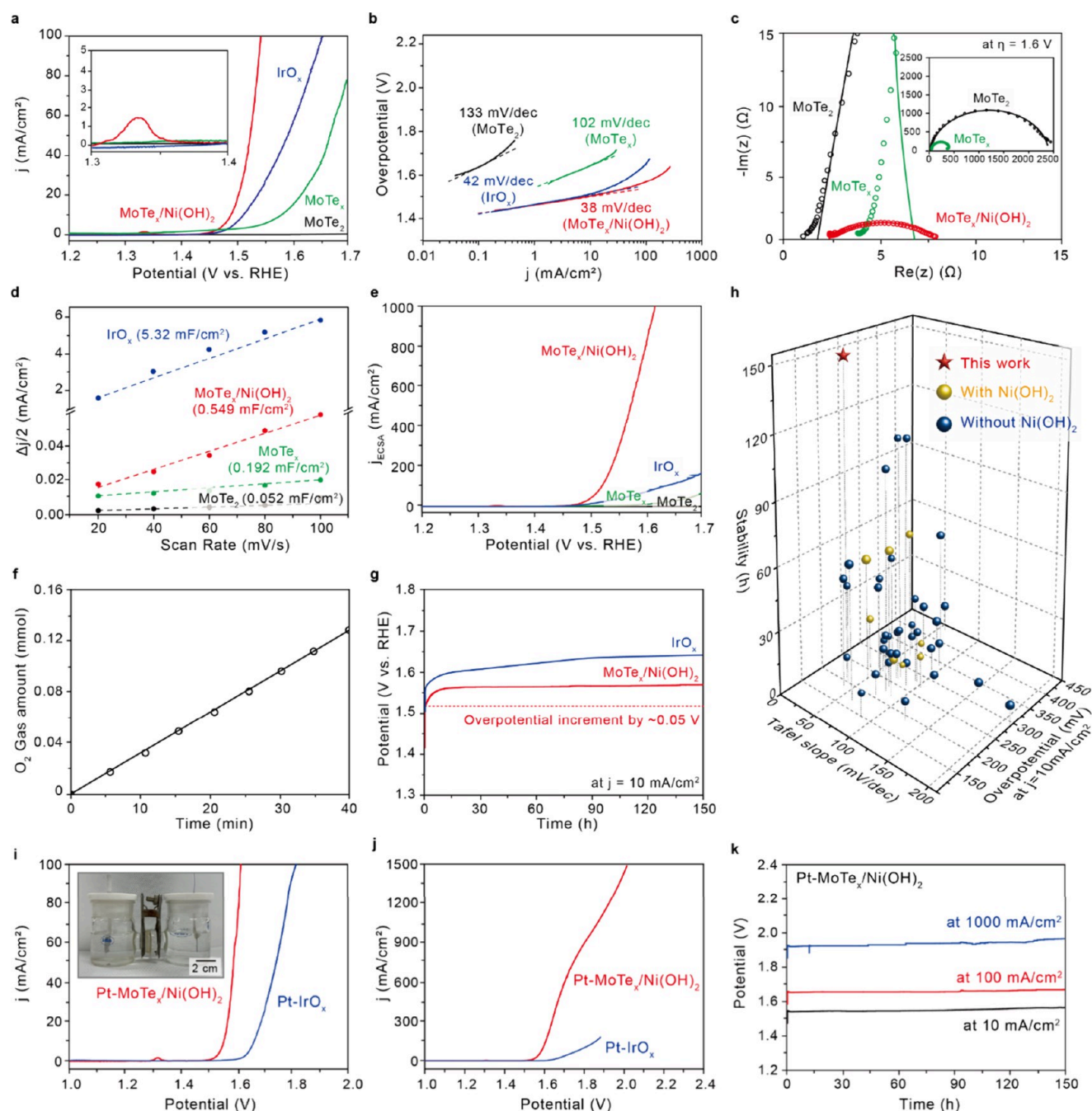


Figure 4. OER activities of MoTe_2 , MoTe_x , and $\text{MoTe}_x/\text{Ni(OH)}_2$ in comparison with IrO_x . (a) iR -corrected polarization curves from MoTe_2 , MoTe_x , $\text{MoTe}_x/\text{Ni(OH)}_2$, and IrO_x and (b) corresponding Tafel plots obtained from the curves. (c) Nyquist plots of the electrochemical cells with the same samples. (d) Determination of electrical double-layer capacitance (C_{DL}) by cyclic voltammetry (CV) and (e) polarization curves normalized to the ECSA estimated in panel (d) – i.e., normalized current density to ECSA (j_{ECSA}) versus potential. (f) Comparison between the amount of collected and theoretical gaseous product (O_2) by the electrochemical cell containing $\text{MoTe}_x/\text{Ni(OH)}_2$ to calculate Faradaic efficiency (Methods). (g) Chronopotentiometry measurements of $\text{MoTe}_x/\text{Ni(OH)}_2$ and IrO_x at $j = 10$ mA/cm 2 for 150 h. (h) OER performance of $\text{MoTe}_x/\text{Ni(OH)}_2$ in comparison to other TMD-based electrocatalysts with and without Ni(OH)_2 . The Δj values were calculated at the open-circuit voltage and R^2 values for all linear regressions are larger than 0.99. In panel (e), ECSA was estimated as follows: $\text{ECSA} \approx C_{DL}/C_s$, where C_s is the specific capacitance of 40 $\mu\text{F}/\text{cm}^2$ for MoTe_2 and 60 $\mu\text{F}/\text{cm}^2$ for IrO_x .^{8,62} In panel (h), each dot has a reference notation corresponding to a Supporting Reference number (No.) (Figure S19). I - V curves of overall water splitting in the electrolyzer over (i) low and (j) high current density regimes. The electrolyzer containing IrO_x as an anode was used as a comparison group in panels (i) and (j). (k) The stability of water splitting from the $\text{MoTe}_x/\text{Ni(OH)}_2$ -based electrolyzer using chronopotentiometry at 10, 100, and 1000 mA/cm 2 .

We experimentally quantified how the defect-induced local phase transitions and formation of Ni(OH)_2 domains enhance the alkaline OER activity of layered MoTe_2 (Figure 4). From the measured polarization curves, the activity of as-transferred MoTe_2 is almost negligible with an onset overpotential (at low

current density, j , of 0.1 mA/cm 2) of 420 mV, for which the OER activity is similar to that of Au films without MoTe_2 (Figure 4a and Figure S15). The current level at this onset overpotential indicates the intrinsic activity of electrocatalysts under near-equilibrium conditions.⁸ The low efficiency of

TMDs is typically caused by reduced charge transfer kinetics and the ratio of active edge sites (to the volume of the crystals).^{31,32,41} After electrochemical Te dissolution and local phase transitions, the onset overpotential decreased to 310 mV due to the formation of metallic 1T' domains that enhance the OER activity of the flakes by suppressing the large charge transfer resistance of MoTe₂ (Figure 4a and Figure S8). The overpotential at a high current density ($j = 10 \text{ mA/cm}^2$) was also reduced to 380 mV for MoTe_x, which indicates the improved electrocatalytic activity achievable by defect and phase engineering.

Subsequent Ni(OH)₂ formation, resulting in MoTe_x/Ni(OH)₂ heterostructures, further improves the OER performance with a low overpotential of 240 mV at 10 mA/cm² (Figure 4a), and this level of activity was stable over high current regimes up to 1 A/cm² (Figure S16). Here, the subpeak at 1.34 V versus the reversible hydrogen electrode (RHE) indicates the oxidation of Ni ions from Ni²⁺ to Ni³⁺ such that hydroxide ions (OH⁻) adsorbed onto the surfaces of Ni(OH)₂ form intermediate NiO(OH) species.^{42,43} We highlight that this level of OER activity from MoTe_x/Ni(OH)₂ is greater than that of Ir-based oxides with the overpotential to achieve the same j with an electrochemical cell containing IrO_x (on carbon cloth, CC) being measured as 290 mV. The nanoscale corrugations with the hydroxide species from Ni(OH)₂ nanoparticles can lead to enhanced mass transfer (with surface hydrophilicity) that contributes to increased OER activity (Figure S17), together with the enhanced atomic-scale reactivities during sequential electron transfer processes as we demonstrate in the later section on free energy calculations. We note that bare Ni(OH)₂ films electrodeposited on Au/Si show substantially lower OER than MoTe_x/Ni(OH)₂ (Figures S18 and S19). While the catalytic activity of Ni(OH)₂ strongly depends on synthesis, morphology, electrode preparation, and testing conditions, MoTe_x/Ni(OH)₂ exhibits an OER activity superior to that of most metal hydroxides (Table S2). Figure 4b highlights that Tafel slopes based on the corresponding polarization curves decreased from 133 mV/dec for as-transferred MoTe₂ to 102 mV/dec (MoTe_x) to 38 mV/dec (MoTe_x/Ni(OH)₂), which is favorable compared to the measured value of 42 mV/dec for IrO_x. The relatively large Tafel slopes observed for MoTe₂ and MoTe_x suggest that O–O bond formation is the rate-determining step, whereas the much smaller slope for MoTe_x/Ni(OH)₂ indicates a mechanistic shift, with the *OH → *O transformation becoming the rate-determining step.^{44–46} Since our stepwise mechanical exfoliation and transfer processes ensure sample-to-sample consistency, the resulting performance of the OER was also reproducible across multiple batches of fabricated WEs with heterostructures (Figure S20).

To understand the improved catalytic activity, we performed electrochemical impedance spectroscopy (EIS) analysis for three-electrode cells containing WEs with MoTe₂, MoTe_x, and MoTe_x/Ni(OH)₂ (Figure 4c, Figure S21). The charge transfer resistance (R_{CT}) determined from Nyquist plots of MoTe₂ was reduced by more than 500-fold after forming heterostructures of MoTe_x/Ni(OH)₂, which suggests that the electron transfer increases between the electrocatalysts and OH⁻ ions in the electrolyte solution to form the initial intermediate species of NiO(OH). The R_{CT} of as-transferred MoTe₂ was similar to that of Au films, indicating the negligible OER activity from the corresponding polarization curve (Figure S15). Although the

R_{CT} level was suppressed after Te dissolution in MoTe_x containing metallic domains, the resulting OER activity remained significantly lower than that of the heterostructures as this catalytic surface could not provide oxygen intermediates with highly reactive sites, as discussed in the theoretical section. We further performed EIS analysis for MoTe₂/Ni(OH)₂ without having Te defects (and associated phase transitions) and confirmed that the R_{CT} of the cell was much larger than that of MoTe_x/Ni(OH)₂, giving rise to lower OER activity and higher Tafel slope (Figure S22). We attribute the relatively sluggish OER from MoTe₂/Ni(OH)₂ to the smaller amounts of metallic 1T' phase domains and Ni(OH)₂ formation compared to MoTe_x/Ni(OH)₂, as confirmed by XPS analysis.

The electrochemically active surface area (ECSA) for each cell was further estimated using the electrochemical double-layer capacitance (C_{DL}), the value of which was determined by measuring the non-Faradaic capacitive current associated with double-layer charging from the scan-rate dependence of cyclic voltammetry (CV) (Figure 4d, Figure S23, Methods). We observed that the C_{DL} of as-transferred MoTe₂ was monotonically enhanced by ~956% (from 0.052 to 0.549 mF/cm²) with the formation of Te defects and local phase transitions followed by Ni(OH)₂ deposition; meanwhile, the C_{DL} of MoTe₂/Ni(OH)₂ was measured to be 0.086 mF/cm² (Figures S22 and S23). Assuming that the C_{DL} is directly proportional to ECSA,^{8,47} this result confirms that defect-mediated heterostructure formation maximizes the density of catalytically active sites for oxygen evolution. Figure 4e shows the polarization curves normalized to the C_{DL} of the corresponding cells, and the relative configurations of the normalized curves (i.e., the current per active area, j_{ECSA} , as a function of overpotential) matched well with the overall trends in Figure 4a. We highlight that the level of j_{ECSA} from MoTe_x/Ni(OH)₂ increased up to 1000 mA/cm² (at the cell potential of 1.61 V vs RHE), which is far greater than IrO_x (j_{ECSA} of 80 mA/cm² at the same potential). The large gap in the OER activity per active area between MoTe_x/Ni(OH)₂ and IrO_x was recorded because we prepared the WE of IrO_x based on the carbon cloth (CC) having a much larger surface area (with a C_{DL} of 5.32 mF/cm²) than that of the flat surface of Au films (Figure S24).

We confirmed that the Faradaic efficiency of the OER from MoTe_x/Ni(OH)₂ was ~97% (Figure 4f) with this high efficiency being stable for over 150 h with an overpotential increment of only ~0.05 V at $j = 10 \text{ mA/cm}^2$ (Figure 4g, Figure S25). Furthermore, the OER performance, surface morphology, and chemical composition of the electrocatalysts remained unchanged after the stability test (Figure S26). In comparison, the stability of IrO_x was less than that of MoTe_x/Ni(OH)₂ with an overpotential increment of 0.09 V over the same period due to the delamination of the materials from the CC substrate (Figure S24). We highlight that this level of long-term stability from the heterostructures was also possible at a much higher j of 1 A/cm² (Figure S16). Figure 4h and Figure S27 compare the overall OER performance of MoTe_x/Ni(OH)₂ in this work with reported electrocatalysts based on TMDs summarized in Table S3. Although the electrochemical formation of Ni(OH)₂ on the 1T phase of MoS₂ achieved comparable electrocatalytic performance (overpotential of 265 mV for $j = 10 \text{ mA/cm}^2$ with the Tafel slope of 35 mV/dec), the stability is limited to 24 h.¹⁵ The growth of NiFe layered double hydroxides on heterophase MoS₂ consisting of 2H and 1T domains exhibited improved OER performance

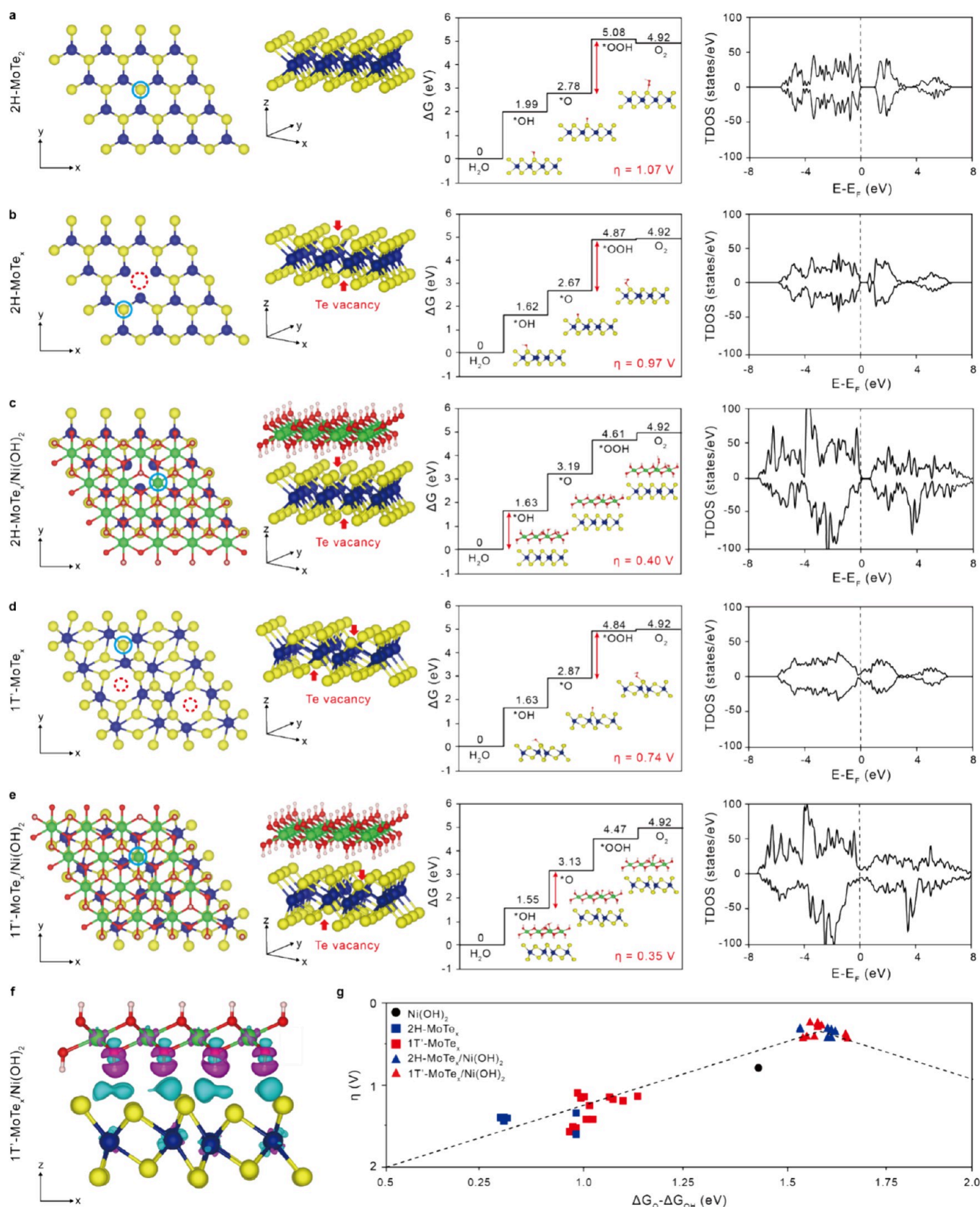


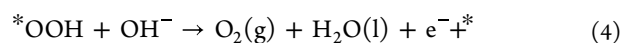
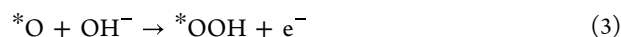
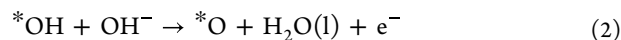
Figure 5. MLP and DFT calculations to understand and quantify atomic-scale OER activity. MLP models to calculate Gibbs free energy diagrams for OER processes and DFT calculations of TDOS for (a) 2H-MoTe₂, (b) 2H-MoTe_x, (c) 2H-MoTe_x/Ni(OH)₂, (d) 1T'-MoTe_x, and (e) 1T'-MoTe_x/Ni(OH)₂. (f) Bader charge analysis of 1T'-MoTe_x/Ni(OH)₂. (g) Volcano plot to compare all the active sites from the models in panels (a)–(e). The free energy diagrams in each model were calculated based on the representative atomic sites marked with cyan circles. In panel (f), regions with electron accumulation/deficiency were marked with cyan/magenta color.

(overpotential of 190 mV for $j = 10 \text{ mA/cm}^2$ with the Tafel slope of 31 mV/dec), but the resulting stability is still only 45 h; these comparisons suggest that the existence of chalcogen defects in the formation processes of these heterostructures influence the durability of hybrid electrocatalysts.²⁰ The growth of 2H-phase MoS_2 on cobalt foams exhibited relatively stable OER (overpotential of 270 mV for $j = 10 \text{ mA/cm}^2$ with the Tafel slope of 74 mV/dec) that last for 100 h.⁴⁸ The fabrication processes for these heterogeneous catalysts, however, contain atomic layer deposition processes that require vacuum processing that limits manufacturing scalability and cost-effectiveness.

Table S4 highlights the performance of our catalyst relative to first-row transition metal oxides or (oxy)hydroxides, such as NiFeO_x . Despite their state-of-the-art activity—sometimes even exceeding the OER activity of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ reported in this work—NiFe-based catalysts are prone to Fe dissolution under sustained anodic polarization, resulting in structural instability and irreversible performance loss during long-term operation.^{49–52} Moreover, the synthesis of high-performing NiFe-based catalysts often involves multistep procedures and stringent compositional control,^{53,54} which may hinder reproducibility and scalability. In contrast, our fabrication strategy—comprising simple mechanical exfoliation followed by electrochemical treatment—relies on only microgram-scale material per electrode yet achieves high electrocatalytic activity coupled with excellent durability. Importantly, the electrochemical Te dissolution step removes only a small fraction ($\sim 8\%$) of Te in a well-controlled aqueous environment without producing harmful volatile species. Collectively, these features underscore the practicality and scalability of our approach.

Toward this end, we constructed a laboratory-scale two-electrode electrolyzer cell based on our $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures to verify the enhanced OER performance for water electrolysis (Figures 4i–k, Figure S28). This full cell consists of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ as the anode and Pt foil as the cathode (denoted as the Pt– $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ cell), with a bipolar membrane employed to prevent Pt contamination of the WE. This cell exhibited a cell potential of 1.54 V at a current density of 10 mA/cm^2 , which is lower than the potential of 1.65 V for a Pt– IrO_x cell (i.e., a comparison system composed of Pt foil as a cathode and IrO_x/CC as an anode), indicating that the 2H–1T' phase boundary and the multilayer structures with $\text{Ni}(\text{OH})_2$ are indeed critical for improved water splitting. We observed that the performance gap between these two cells is significant, particularly in the high current density regime with the Pt– $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ cell demonstrating a current density of up to $\sim 1500 \text{ mA/cm}^2$ at a potential of 2.2 V, while only 12% of the current density level (i.e., $\sim 185 \text{ mA/cm}^2$) was generated for the Pt– IrO_x cell at the same potential (Figure 4j). Current density levels exceeding 1000 mA/cm^2 suggest that the system can be utilized for the mass production of molecular hydrogen for industrial-scale applications. We also verified that such a level of overall water splitting with the heterostructures was achievable by using different cell configurations (Figure S29). We further characterized the long-term stability of the electrolyzer containing $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ by chronopotentiometry, highlighting the long-term stability at low and high current levels with the electrolyzer maintaining current levels with negligible performance degradation for 150 h of operation (j of 10, 100, and 1000 mA/cm^2 at 1.54, 1.62, and 1.91 V, respectively).

We also explored the mechanisms of the OER enhancement for MoTe_x and $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures using machine-learning potential (MLP) calculations (Methods, Figure 5, Figure S30). Here, we considered the conventional four-step processes in OER through the formation of $^*\text{OH}$, $^*\text{O}$, and $^*\text{OOH}$ intermediates as follows:



We checked that the rate-determining step (RDS) for the OER on pristine 2H– MoTe_2 was step (3) with the highest Gibbs free energy barrier (ΔG_3) of 2.3 eV, indicating that the theoretically required OER overpotential (η) is 1.07 V (Methods, Figure 5a). This value of η was slightly reduced to 0.97 V for 2H– MoTe_x with the same RDS, as suggested by the corresponding Tafel slopes in Figure 4b. The calculated ΔG to form $^*\text{OOH}$ on the Te atom adjacent to the vacancy was 2.2 eV (Figure 5b and Figure S31). We calculated that η of 2H– $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ was significantly decreased to 0.4 V with the Ni atoms situated close to the underlying Te vacancy mostly contributing to the overall OER performance (Figure 5c and Figure S32). We note that η on Ni atoms from a bare $\text{Ni}(\text{OH})_2$ model was 0.69 V with the corresponding RDS being step (1). This RDS with a barrier energy (ΔG_1) of 1.92 eV was lowered by forming heterostructures since the presence of defective MoTe_x facilitates the deprotonation on $\text{Ni}(\text{OH})_2$.¹⁸ Furthermore, Bader charge analysis of the heterostructures by density functional theory (DFT) exhibits electron transfer from $\text{Ni}(\text{OH})_2$ to 2H– MoTe_x , inducing hole accumulation on the metal hydroxides that accelerates the adsorption of OH^- and the electron transfer from OH^- to form active $^*\text{OH}$, thus enhancing overall OER kinetics. This simulated local electron transfer is also consistent with the aforementioned XPS measurements (Methods, Figures S33 and S34).

In comparison to 2H– MoTe_x , 1T'– MoTe_x enables higher OER reactivity with the same RDS (ΔG_3 of 1.97 eV), resulting in η of 0.74 V (Figure 5d and Figure S35). Figure 5e and Figure S36 highlight that the heterostructure consisting of 1T'– MoTe_x and $\text{Ni}(\text{OH})_2$ creates the most reactive atomic sites. We theoretically found that η as low as 0.35 V is achievable on the exterior Ni atoms in the 1T'– $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ model in which both step (1) and step (2) could serve as the RDS; this theoretical interpretation is fully consistent with the experimental observation of a significantly reduced Tafel slope. Here, the atomic-scale reactivity to form the $^*\text{O}$ intermediate (ΔG_2 of 1.58 eV) was quantitatively similar to the reactivity for $^*\text{OH}$ (ΔG_1 of 1.55 eV) in the four-electron transfer processes. Considering the theoretical η values of 0.4–0.6 eV for IrO_2 from the reported literature,^{55–58} our experimental OER performance of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ exceeding that of IrO_x can be attributed to both constituent 2H– $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ and 1T'– $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ domains. It is noteworthy that bare $\text{Ni}(\text{OH})_2$ exhibits the higher overpotential of 0.69 V (Figure S37) compared to the heterostructures, consistent with the experimental comparison shown in Figure S19. Compared with bare $\text{Ni}(\text{OH})_2$, the heterostructure exhibits stronger electron donation from MoTe_x to interfacial Ni sites, stabilizing the adsorbed intermediates during the OER (Figure

S38). Based on calculated charge redistributions (CR), we extracted the $\text{CR}_\text{O}-\text{CR}_\text{OOH}$ parameter, which correlates linearly with the overpotential.⁵⁹ The $\text{CR}_\text{O}-\text{CR}_\text{OOH}$ value of the heterostructure was only $\sim 55\%$ of that of bare $\text{Ni}(\text{OH})_2$, reducing the energy penalty of the potential-limiting step and thereby lowering η by $\sim 50\%$.

The charge distribution analysis also supports the electron transfer from $\text{Ni}(\text{OH})_2$ to $1\text{T}'\text{-MoTe}_x$ to improve the resulting OER, which is a trend that is similar to the case of $2\text{H-MoTe}_x/\text{Ni}(\text{OH})_2$; however, we observed more significant electron deficiency on the metal hydroxides due to the metallic $1\text{T}'$ phase with higher conductivity (Figure S5f). We correlated OER reactivities of all of the atomic sites collected by MLP with the resulting OER activities in a volcano plot (Figure S5g). Here, we derived a scaling factor using theoretical η values corresponding to the $\Delta G_\text{O} - \Delta G_\text{OH}$ values, which serves as an activity descriptor,^{58,60,61} allowing us to conclude that overall OER performance improves progressively from MoTe_2 to MoTe_x , and further to $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures. Notably, when $\Delta G_\text{O} - \Delta G_\text{OH}$ value reaches 1.6 eV, the system exhibits the lowest η , matching well with the predicted theoretical trend.

To investigate the electronic structures of the model catalysts, we calculated the total density of states (TDOS) corresponding to each (hetero)structure before adsorbing oxygen intermediates (Figure S5a–e, right column). We calculated that additional electron energy states within the bandgap of 2H-MoTe_2 by incorporating Te vacancies to form 2H and (metallic) $1\text{T}'\text{-MoTe}_x$ with increased level of electron conductivity. These TDOS trends are further enhanced after forming $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures especially near the Fermi level (E_F), which is expected to improve the electron transfer kinetics in the OER. In particular, we observed a gapless electronic structure for the $1\text{T}'\text{-MoTe}_x/\text{Ni}(\text{OH})_2$ models, which suggests that the increased electron population occupying the energy levels near the band edges support higher OER activity. Although the $1\text{T}'\text{-MoTe}_x$ before $\text{Ni}(\text{OH})_2$ formation exhibits metallic behavior, the resulting OER activity was quantitatively less than those of the heterostructures due to the lower level of overall TDOS. We note that DOS above E_F (i.e., $E - E_\text{F}$ values between 0 and 5 eV) were enhanced by forming the heterostructures in comparison with the electronic structure of bare $\text{Ni}(\text{OH})_2$ with the $1\text{T}'\text{-MoTe}_x$ serving as the reservoir of electrons for the OER in the conduction band regime (Figure S37).

CONCLUSIONS

In summary, we have developed a method for realizing efficient and stable OER electrocatalysts based on earth-abundant elements. Although tellurium is not a precious metal, its relatively low crustal abundance ($\sim 1 \mu\text{g/kg}$) highlights the need for judicious use as a precursor in the synthesis of layered transition metal dichalcogenides. In this work, we electrochemically controlled the limited dissolution of Te from mechanically exfoliated MoTe_2 followed by $\text{Ni}(\text{OH})_2$ deposition, to construct heterostructures comprising $2\text{H-MoTe}_x/\text{Ni}(\text{OH})_2$ and $1\text{T}'\text{-MoTe}_x/\text{Ni}(\text{OH})_2$ domains. Specifically, electrochemical Te dissolution from mechanically exfoliated MoTe_2 followed by $\text{Ni}(\text{OH})_2$ deposition results in heterostructures consisting of $2\text{H-MoTe}_x/\text{Ni}(\text{OH})_2$ and $1\text{T}'\text{-MoTe}_x/\text{Ni}(\text{OH})_2$ domains in which alkaline OER exceeds IrO_x on centimeter-scale electrodes. MLP and DFT calculations evaluated atomic-scale active sites for the heterostructures,

thus revealing the OER enhancement mechanisms based on four-electron transfer processes with reduced energy barriers. Because this process can be applied to other TMDs and metal hydroxides, we anticipate that our work will provide a generalizable strategy for control the electrocatalytic activity for diverse reactions including hydrogen evolution and CO_2 reduction.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c15520>.

Method; Schematic procedure for the mechanical exfoliation and transfer of MoTe_2 ; Experimental procedures for the formation of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ on Au/Si electrodes; Characterization of the thickness and size distributions of as-transferred MoTe_2 flakes; Formation of MoTe_2 -based working electrode for OER; Reproducibility of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures across samples; AFM images of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures across samples; XPS survey spectrum; Quantification of compositional ratio of MoTe_2 , MoTe_x , and $\text{MoTe}_x/\text{Ni}(\text{OH})_2$; Compositional analysis of $\text{MoTe}_2/\text{Ni}(\text{OH})_2$; Compositional comparisons among MoTe_2 , MoTe_x , and $\text{MoTe}_2/\text{Ni}(\text{OH})_2$; Raman signal of the $1\text{T}'$ phase in as-transferred MoTe_2 ; Raman spectra and Raman maps of $\text{MoTe}_2/\text{Ni}(\text{OH})_2$; TEM images of MoTe_2 and MoTe_x ; EDS maps of MoTe_2 and MoTe_x ; OER performance of as-transferred MoTe_2 compared with the Au/Si substrate; OER performance of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ at high current densities; Surface wetting properties of MoTe_2 , MoTe_x , $\text{MoTe}_x/\text{Ni}(\text{OH})_2$, and Au/Si surface; Characterization of electrodeposited $\text{Ni}(\text{OH})_2$ films; Comparison of OER performance between bare $\text{Ni}(\text{OH})_2$ and $\text{MoTe}_x/\text{Ni}(\text{OH})_2$; Electrochemical consistency across batches of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ working electrode cells; EIS measurements of MoTe_2 , MoTe_x , and $\text{MoTe}_x/\text{Ni}(\text{OH})_2$; Comparison of OER performance of MoTe_2 , $\text{MoTe}_x/\text{Ni}(\text{OH})_2$, and $\text{MoTe}_2/\text{Ni}(\text{OH})_2$; Measurements of non-Faradaic capacitance current to estimate C_DL ; SEM image of IrO_x/CC electrodes; Long-term stability of $\text{MoTe}_2/\text{Ni}(\text{OH})_2$; Characterization of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures before and after long-term stability test; Comparisons of OER performances of Mo-based TMDs electrocatalysts; Schematic illustration of a two-electrode electrolyzer system; Two-electrode beaker-type electrolyzer containing alkaline electrolyte; Model structures of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$; Gibbs free energy diagram of 2H-MoTe_x ; Gibbs free energy diagram of $2\text{H-MoTe}_x/\text{Ni}(\text{OH})_2$; Bader charge analysis of $2\text{H-MoTe}_x/\text{Ni}(\text{OH})_2$; XPS binding energy shifts by the formation of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$; Gibbs free energy diagram of $1\text{T}'\text{-MoTe}_x$; Gibbs free energy diagram of $1\text{T}'\text{-MoTe}_x/\text{Ni}(\text{OH})_2$; Gibbs free energy diagram of $\text{Ni}(\text{OH})_2$; Interfacial charge redistribution occurring in of $\text{Ni}(\text{OH})_2$ and $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ during OER; Statistical analysis of $\text{MoTe}_x/\text{Ni}(\text{OH})_2$ heterostructures across samples; Comparisons of OER performances of bare $\text{Ni}(\text{OH})_2$ electrocatalysts; Comparisons of OER performances of Mo-based TMDs electrocatalysts; Comparisons of OER performances of first-row transition metal (oxy)-

hydroxides electrocatalysts; Supporting Information references (PDF)

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Notes

The authors declare no competing financial interest.

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